

Final Report ILO 3

Legal Assessment



CREATING MEANINGFUL EXPERIENCES

Applied Data Science & AI

Date

Index

Task 2.1: Identification of The Level of Risk and GDPR Implications (Individual Task)	2
Task 2.2: AI System Development: Legal Risk Assessment. (Group Task)	4
References	7

Task 2.1: Identification of The Level of Risk and GDPR Implications (Individual Task)

1.1 Proposed AI system

The proposed system is an intelligent system that detects and predicts short-term market anomalies like crashes, dips, and rallies. It would be an early warning tool for traders, analysts, hedge funds, or other financial platforms. The main sector where the tool operates is the finance sector, especially in the market analysis and trading intelligence subsector. The following key functionalities can be distinguished:

- Ingests and processes minute-by-minute OHLCV (Open, High, Low, Close, Volume) data
- Applies feature engineering to create relevant indicators (e.g., moving averages, volatility measures)
- Uses classification models to identify market events
- Implements anomaly detection for flash crashes/spikes
- Optional: Incorporates NLP for news sentiment analysis to enhance predictions
- Provides alerts through a user interface when specific market events are detected
- Offers visualization through a dashboard
- Stores data and model outputs in PostgreSQL database

1.2 The level of risk

The proposed tool falls under the “Limited Risk” category of the EU AI Act Risk Classification. This classification is justified by eliminating other risk categories that do not apply.

The system does not meet the criteria for the “Unacceptable Risk” category, as it does not engage in social scoring, manipulative behavior, or the exploitation of vulnerable population groups. It also does not fall under the “High-Risk” classification, as it is not a biometric identification system, does not manage critical infrastructure, and is not used in education, employment, law enforcement, or other regulated high-risk domains.

At the same time, the project exceeds the scope of “Minimal Risk”, as it operates in the financial sector, where users’ decisions may have significant economic consequences. The tool provides predictive insights that could influence investment choices, making the impact more substantial than what is expected from minimal-risk applications.

Therefore, through a process of elimination, the proposed AI system is most appropriately classified as a Limited Risk system.

1.3 Legal obligations and requirements

Applicable Legal Obligations:

- **Transparency Requirements (Article 50)**
 - **User Notification:** Users must be clearly informed that they are interacting with an AI system.
 - **Output Labeling:** All predictions must be explicitly labeled as AI-generated.
- **AI Literacy (Article 4)**
 - AI providers and deployers must ensure that all relevant staff and individuals involved in the development or use of the system possess an adequate level of AI literacy. This should be tailored to their technical background, training, and the specific context and intended users of the system.

Excluded Legal Requirements:

- High-Risk Compliance
- Prohibited Practice Safeguards

Reasoning for Inclusion of Obligations

According to the compliance guidance provided by the drafters of the EU AI Act, only Article 4 and Article 50 are applicable to the proposed system. These include the obligation to inform users that they are interacting with an AI system, and to clearly label outputs as AI-generated.

This requirement is rooted in the principle of transparency, as users may otherwise mistake AI-generated content for human judgment, like chatbot scenarios. Therefore, while high-risk and prohibited practice safeguards are not relevant to this tool, transparency and literacy requirements must be observed.

Rationale Based on Authoritative Sources

The classification of the proposed AI tool as a Limited Risk system is based on a careful review of the EU Artificial Intelligence Act (2024), particularly the definitions and examples outlined in Annex III (High-Risk Systems) and Article 5 (Prohibited Practices). The tool does not fall under these categories due to its domain of use and its functionalities, which align more closely with the examples of Limited Risk systems discussed in official EU briefings and legal analyses (e.g. AI chatbots, content recommendation engines).

Additionally, the transparency and literacy obligations are consistent with the output of the EU AI Act Compliance Checker, a tool designed to help organizations assess their legal responsibilities under the Act. While not legally binding, the checker translates the regulation into actionable guidance based on the system's characteristics. In this case, it identified Article 4 (AI Literacy) and Article 50 (Transparency) as the only applicable obligations, confirming that the tool must clearly inform users of AI involvement and label its predictions accordingly. This aligns with guidance from regulatory bodies such as the European Commission and supports the classification of the tool as Limited Risk, requiring only specific transparency measures.

Task 2.2: AI System Development: Legal Risk Assessment. (Group Task)

1.3 Developed AI system

The proposed system is an intelligent system that detects and predicts short-term market anomalies like crashes, dips, and rallies. It would be an early warning tool for traders, analysts, hedge funds, or other financial platforms. The main sector where the tool operates is the finance sector, especially in the market analysis and trading intelligence subsector. The following key functionalities can be distinguished:

- Ingests and processes minute-by-minute OHLCV (Open, High, Low, Close, Volume) data
- Applies feature engineering to create relevant indicators (e.g., moving averages, volatility measures)
- Uses classification models to identify market events
- Implements anomaly detection for flash crashes/spikes
- Optional: Incorporates NLP for news sentiment analysis to enhance predictions
- Provides alerts through a user interface when specific market events are detected
- Offers visualization through a dashboard
- Stores data and model outputs in PostgreSQL database

1.4 The level of risk

The proposed tool falls under the “Limited Risk” category of the EU AI Act Risk Classification. This classification is justified by eliminating other risk categories that do not apply.

The system does not meet the criteria for the “Unacceptable Risk” category, as it does not engage in social scoring, manipulative behavior, or the exploitation of vulnerable population groups. It also does not fall under the “High-Risk” classification, as it is not a biometric identification system, does not manage critical infrastructure, and is not used in education, employment, law enforcement, or other regulated high-risk domains.

At the same time, the project exceeds the scope of “Minimal Risk”, as it operates in the financial sector, where users' decisions may have significant economic consequences. The tool provides predictive insights that could influence investment choices, making the impact more substantial than what is expected from minimal-risk applications.

Therefore, through a process of elimination, the proposed AI system is most appropriately classified as a Limited Risk system.

2.3 Legal obligations and requirements

Applicable Legal Obligations:

- **Transparency Requirements (Article 50)**
 - **User Notification:** Users must be clearly informed that they are interacting with an AI system.
 - **Output Labeling:** All predictions must be explicitly labeled as AI-generated.
- **AI Literacy (Article 4)**
 - AI providers and deployers must ensure that all relevant staff and individuals involved in the development or use of the system possess an adequate level of AI literacy. This should be tailored to their technical background, training, and the specific context and intended users of the system.

Excluded Legal Requirements:

- High-Risk Compliance
- Prohibited Practice Safeguards

Reasoning for Inclusion of Obligations

According to the compliance guidance provided by the drafters of the EU AI Act, only Article 4 and Article 50 are applicable to the proposed system. These include the obligation to inform users that they are interacting with an AI system, and to clearly label outputs as AI-generated.

This requirement is rooted in the principle of transparency, as users may otherwise mistake AI-generated content for human judgment, like chatbot scenarios. Therefore, while high-risk and prohibited practice safeguards are not relevant to this tool, transparency and literacy requirements must be observed.

Rationale Based on Authoritative Sources

The classification of the proposed AI tool as a Limited Risk system is based on a careful review of the EU Artificial Intelligence Act (2024), particularly the definitions and examples outlined in Annex III (High-Risk Systems) and Article 5 (Prohibited Practices). The tool does not fall under these categories due to its domain of use and its functionalities, which align more closely with the examples of Limited Risk systems discussed in official EU briefings and legal analyses (e.g. AI chatbots, content recommendation engines).

Additionally, the transparency and literacy obligations are consistent with the output of the EU AI Act Compliance Checker, a tool designed to help organizations assess their legal responsibilities under the Act. While not legally binding, the checker translates the regulation into actionable guidance based on the system's characteristics. In this case, it identified Article 4 (AI Literacy) and Article 50 (Transparency) as the only applicable obligations, confirming that the tool must clearly inform users of AI involvement and label its predictions accordingly. This aligns with guidance from regulatory bodies such as the European Commission and supports the classification of the tool as Limited Risk, requiring only specific transparency measures.

2.4 Compliance

The AI system, designed for detecting and predicting short-term market anomalies and classified as "Limited Risk" under the EU AI Act, will adhere to its legal obligations by implementing specific measures focused on AI literacy (Article 4) and transparency (Article 50).

Compliance with Article 4 (AI Literacy)

- To ensure an adequate level of AI literacy for all individuals involved with the system, including internal teams and external investors, the following measures will be implemented:
- **Comprehensive AI Literacy Programs and Resources:** Implement tailored training programs for internal staff (developers, analysts) covering the AI system's technology, operational capabilities, and limitations. For external investors, provide accessible educational resources, including clear documentation on the system's functionality, data sources, algorithms in non-technical terms, and specific training modules on interpreting AI outputs responsibly.
- **Transparent Communication on System Capabilities and Limitations:** Clearly communicate to all users, especially external investors, that the AI system is a supportive tool for market analysis and not a replacement for human judgment or definitive financial advice. This includes explicit disclosures about the system's inherent limitations, such as its inability to predict all market events or account for non-quantifiable factors, and outlining potential risks associated with its use.
- **Ongoing Support, Documentation, and Updates:** Maintain and provide access to comprehensive and up-to-date documentation for all user groups. Offer continuous support channels for users to ask questions and receive assistance. Regularly inform all users, including investors, about significant updates or changes to the AI system, its algorithms, data sources, or risk assessments to ensure sustained understanding and responsible use.

Compliance with Article 50 (Transparency Obligations)

- To meet transparency requirements and ensure users are clearly informed that they are interacting with an AI system and that its outputs are AI-generated, the following measures will be integrated into the system's design and user interface:
- **Clear User Notification:** The system's user interface (UI), including any dashboards that display market analyses or alerts, will feature a prominent and easily understandable notification. This notification will explicitly state that the insights, predictions (e.g., for market crashes, dips, rallies), and alerts are generated by an artificial intelligence system.
- **Explicit Labeling of AI Outputs:** All predictions, such as detected market anomalies, and any subsequent alerts or visualizations provided through the UI or other communication channels will be clearly and unambiguously labeled (e.g., "AI-Generated Prediction," "AI Market Analysis"). This practice is crucial to prevent users from mistaking AI outputs for human-derived analysis or definitive financial advice.
- **Understandable Explanations (Where Feasible):** While the inherent complexity of certain AI models might limit the provision of exhaustive explanations for every prediction, the system will endeavor to offer users insights into the key factors or indicators that contributed to a particular alert or prediction, where technically feasible. This approach aims to enhance user understanding, trust, and the ability to critically evaluate the AI's outputs.

These compliance measures are designed to ensure the responsible development and deployment of the AI system. They align with the "Limited Risk" classification by prioritizing user awareness, informed decision-making, and understanding of the AI's role, thereby mitigating risks associated with the misinterpretation or over-reliance on AI-driven financial insights.

References

Article 4: AI literacy | *EU Artificial Intelligence Act*. (n.d.). Retrieved May 11, 2025, from <https://artificialintelligenceact.eu/article/4/>

Article 50: Transparency Obligations for Providers and Deployers of Certain AI Systems | *EU Artificial Intelligence Act*. (n.d.). Retrieved May 11, 2025, from <https://artificialintelligenceact.eu/article/50/>

EU AI Act Compliance Checker | *EU Artificial Intelligence Act*. (n.d.). Retrieved May 11, 2025, from <https://artificialintelligenceact.eu/assessment/eu-ai-act-compliance-checker/>

EU AI Act risk groups | *Autoriteit Persoonsgegevens*. (2025, April 9).

<https://www.autoriteitpersoonsgegevens.nl/en/themes/algorithms-ai/eu-ai-act/eu-ai-act-risk-groups>

EU AI Act: Risk-Classifications of the AI Regulation. (n.d.). Retrieved May 11, 2025, from <https://www.trail-ml.com/blog/eu-ai-act-how-risk-is-classified>



Games



Leisure & Events



Tourism



Media



Data Science & AI



Hotel



Logistics



Built Environment



Facility

Mgr. Hopmansstraat 2
4817 JS Breda

P.O. Box 3917
4800 DX Breda
The Netherlands

PHONE

+31 76 533 22 03

E-MAIL

communications@buas.nl

WEBSITE

www.BUas.nl

CREATING MEANINGFUL EXPERIENCES